ALPHA BOY'S

AICTE | IBM SkillsBuild Gen AI & Cloud Computing Internship | BharatCares

Team Details

Unique Team ID	IBMBH03605
College Pincode	395007
College Name	C.K. Pithawala College of Engineering and Technology, Surat.
Internship Name	AICTE | IBM SkillsBuild Gen AI & Cloud Computing Internship | BharatCares
Total Team Members	05
Total Female Team Members	00
Team Leader Name	Patil Devendra Vinod (230090116054)
Team Leader Email	patildevendra984@gmail.com
Team members name	• Daksh Bhavnani (230090116012)
	• Dhruv S. Maisuria (230090116016)
	• Tilak Parmar (230090116041)
	• Rana Hemraj (230090116057)

PROJECT REPORT

Bharat Cares & IBM — GenAI Application Development Program

1. Application Overview & Tech Stack

Job seekers in today's automated hiring landscape face two compounding problems. First, the ATS problem: candidates are silently rejected by Applicant Tracking Systems due to weak formatting or missing contextual keywords — often before a human reviewer ever sees their resume. Second, the workflow problem: career-preparation tools are siloed — a candidate builds a resume on one site, writes a cover letter on another, and practices interviews on a third, with no shared profile context across any of them.

ResumeAI — Smart Career Studio solves both by unifying every stage of job preparation into a single authenticated workspace. The platform carries the candidate's profile context from resume building and AI-driven bullet enhancement, through semantic ATS compatibility scoring and automated cover letter generation, into a timed aptitude screening round and a fully adaptive AI mock interview — without the candidate ever switching tools.

A university graduate targeting a "Python Developer" role, for example, can build an ATS-friendly resume in the live split-screen builder, refine weak bullet points with the AI Bullet Enhancer, check their Compatibility Score against the actual job description, generate a tailored cover letter for their target company, complete a 20-question timed aptitude quiz, and sit a ·:35-minute adaptive AI mock interview — all within one session on one platform.

Target Users

- Job Seekers / Candidates — to build ATS-optimised resumes, identify specific skill gaps against a real job description, rehearse aptitude tests under timed conditions, and practice adaptive technical interviews before facing them live.
- Recruiters / HR Admins — to review proctored assessment reports, ATS compatibility scores, aptitude results, and AI-generated qualitative hiring summaries per candidate, reducing manual screening time.

Core Feature Modules

- User Authentication (My Career Space): secure registration and login with a persistent 'Remember me for 30 days' session; saved resume drafts, job-application tracker, and persistent interview history per account.
- Live Resume Builder: split-screen editor — categorised input tabs (Basics, Summary, Experience, Education, Skills) with word counter and completion tracker on the left; a live white-canvas document preview, selectable templates and accent colours, and one-click PDF export on the right.
- Dual-Mode AI Assistant: Career Intake Coach — a conversational chat that organically maps out the user's background and skills; and a Bullet Enhancer that transforms a rough draft bullet into a metric-driven professional achievement with one-click copy.
- Contextual ATS Analyzer: upload a resume PDF (up to 5 MB) alongside a pasted job description; returns a semantic Compatibility Score with a structured Matched Skills / Missing Skills breakdown powered by ChatGPT NLP.
- Automated Cover Letter Generator: contextualises the user's resume data against a specified company and role via ChatGPT, outputting a fully formatted, editable draft directly in an adjacent text field.
- Aptitude Quiz Simulator (Stage 1 Screening): 20 randomised MCQs on a chosen domain under a strict 20-minute countdown; local answer autosave; forced auto-submit at time-up; anti-cheat tab-refresh warnings.
- Adaptive AI Mock Interview (Stage 2 Screening): ·:35-minute conversational interview driven by Gemini, keyed to the candidate's saved profile and latest ATS score. Gemini acts as the hiring manager and dynamically adapts behavioural and technical questions to the target job role. Background proctoring logs tab-switch and window-blur events.

- Personalised Learning Roadmap: compares quiz and interview scores against role benchmarks; generates a 4-week study calendar with curated free-course links (Harvard CS50, MIT OCW, and others).
- Career Matcher: generates direct, profile-matched job-search links to LinkedIn, Indeed, and Naukri for the candidate's target role and location.

1.1 Tech Stack

Layer	Technology	Reason Selected
Programming Language	Python 3.11	Rich standard libraries, native AI SDK integrations, and robust scripting for Flask middleware and PDF text extraction.
Web Framework	Flask 3.0.3	Lightweight, route-based, and extensible; minimal runtime footprint with no compilation pipeline.
Frontend	HTML5, Vanilla CSS3, JavaScript	Async fetch() calls prevent page reloads, enabling the live resume preview. No framework overhead; fast load times on low-bandwidth connections.
Database	SQLite3 with mysql_store proxy adapter	Zero external dependency; 10-table relational schema with cascade deletes. The mysql_store.py emulation layer means all routes use MySQL-compatible syntax, enabling a connector-swap migration to hosted MySQL/ PostgreSQL with no route rewrites.
AI — Primary	Google Gemini API (gemini-3.5-flash)	Speed and conversational fluidity make it the right choice for real-time turn-by-turn modules: Career Intake Coach, Bullet Enhancer, and the AI Mock Interview.
AI — Secondary	OpenAI ChatGPT / OpenRouter Gateway	Deeper semantic reasoning and structural adherence produce higher-quality ATS scores and cover letter drafts. OpenRouter provides a failover path to GPT-4o-mini, Anthropic, or Groq when primary keys are rate-limited.
PDF Parsing	pypdf / pymupdf	Reliable text extraction from candidate-uploaded resume PDFs for ATS scoring; handles text-based PDFs up to 5 MB.
Production Server	Gunicorn	High-performance WSGI server for concurrent worker thread management on Unix/Linux deployment targets.
Design System	Dark glassmorphism — neon purple & teal accents	Polished, low-eye-strain aesthetic optimised for long career-preparation sessions; consistent across all seven modules via shared CSS variables.
Version Control	Git & GitHub	Distributed version control with deployment hooks to Render; peer-review cycle across the 6-person team.

2. AI Integration

ResumeAI uses a deliberate dual-LLM routing architecture, pairing each module with the provider whose strengths best match its task profile. This avoids the cost and latency of routing every call through a single general-purpose model, while the fallback chain ensures platform availability when any single provider hits a rate limit or billing threshold.

2.1 Model Routing by Module

Module	Provider	Rationale
Career Intake Coach	Google Gemini (gemini-3.5-flash)	Conversational fluidity and low latency are critical for a multi-turn chat that maps user background organically.
Bullet Enhancer	Google Gemini (gemini-3.5-flash)	Rapid single-turn text transformation; Gemini's speed keeps the copy-edit loop instant.
Adaptive AI Mock Interview	Google Gemini (gemini-3.5-flash)	Dynamically adapts behavioural and technical questions turn-by-turn; requires fast, context-aware generation under a real-time pacing timer.
Contextual ATS Analyzer	OpenAI ChatGPT (via OpenRouter)	Deep NLP comparison between resume and job description; ChatGPT's semantic precision produces more reliable Matched / Missing Skills categorisation than a speed-optimised model.
Cover Letter Generator	OpenAI ChatGPT (via OpenRouter)	Strict professional formatting and structural adherence are essential for a document the user will send to an employer.

2.2 Fallback Chain

Every AI call passes through a priority-ordered provider chain. If a provider returns a 402, 404, or network timeout, the application logs a warning and automatically retries the next provider without surfacing an error to the user:

- 1st — Local OpenAI API key credentials
- 2nd — Local Gemini API key credentials (targeting gemini-3.5-flash)
- 3rd — OpenRouter external gateway endpoint
- 4th — Groq / xAI developer keys

2.3 Prompting Strategy

Prompts are structured as isolated system instruction + candidate message context pairs. The system instruction defines the AI's role and output contract (format, fields, constraints), while the message context supplies the candidate's resume, target role, and — for the mock interview — the growing conversation history. This separation keeps utility calls (bullet enhancement, cover letter) cheap and fast, while semantic calls (ATS scoring) receive richer context without bloating simpler flows.

For mock interview answer grading, the model receives the exact question text, a list of expected technical concepts, and the candidate's verbatim answer. The instruction mandates a structured output: a numerical score (0–100), a list of missing concepts, and a short diagnostic feedback string. The ATS Analyzer similarly returns a compatibility percentage plus two categorised lists rather than a free-form summary. All max_tokens values are explicitly capped per request to control cost and prevent Render free-tier credit exhaustion.

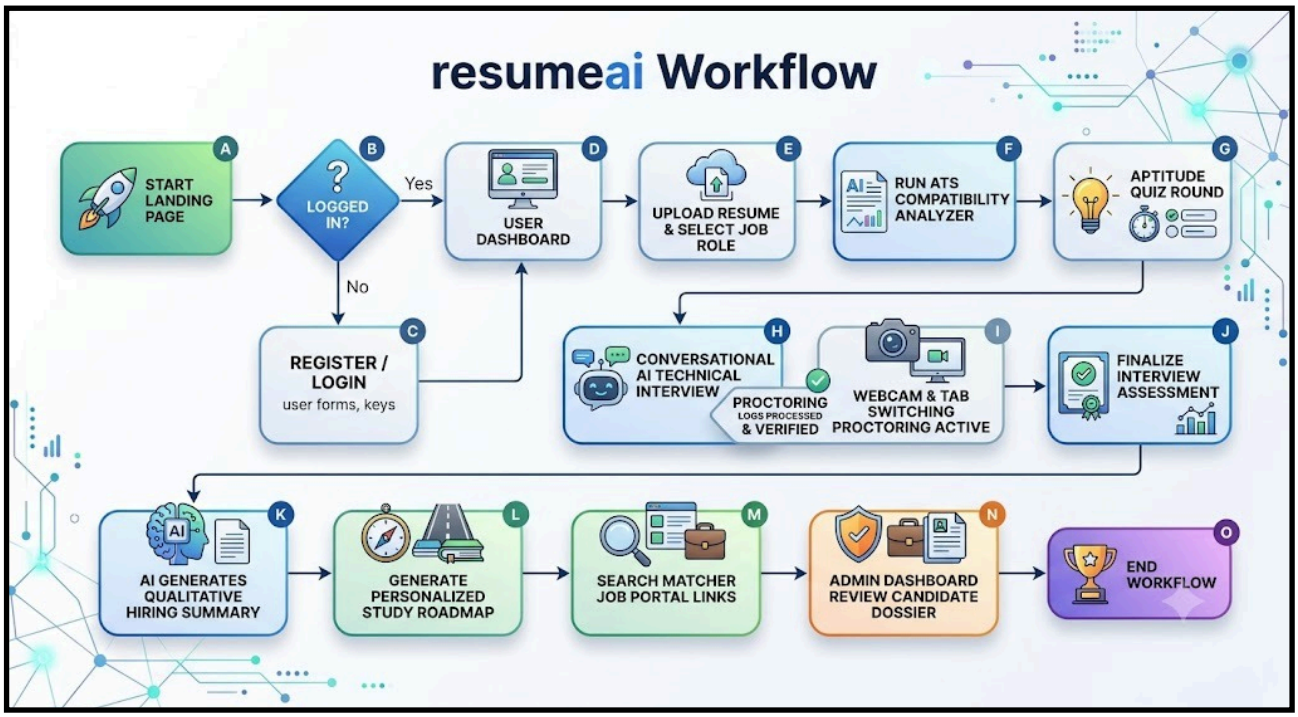
3. Phase - By - Development

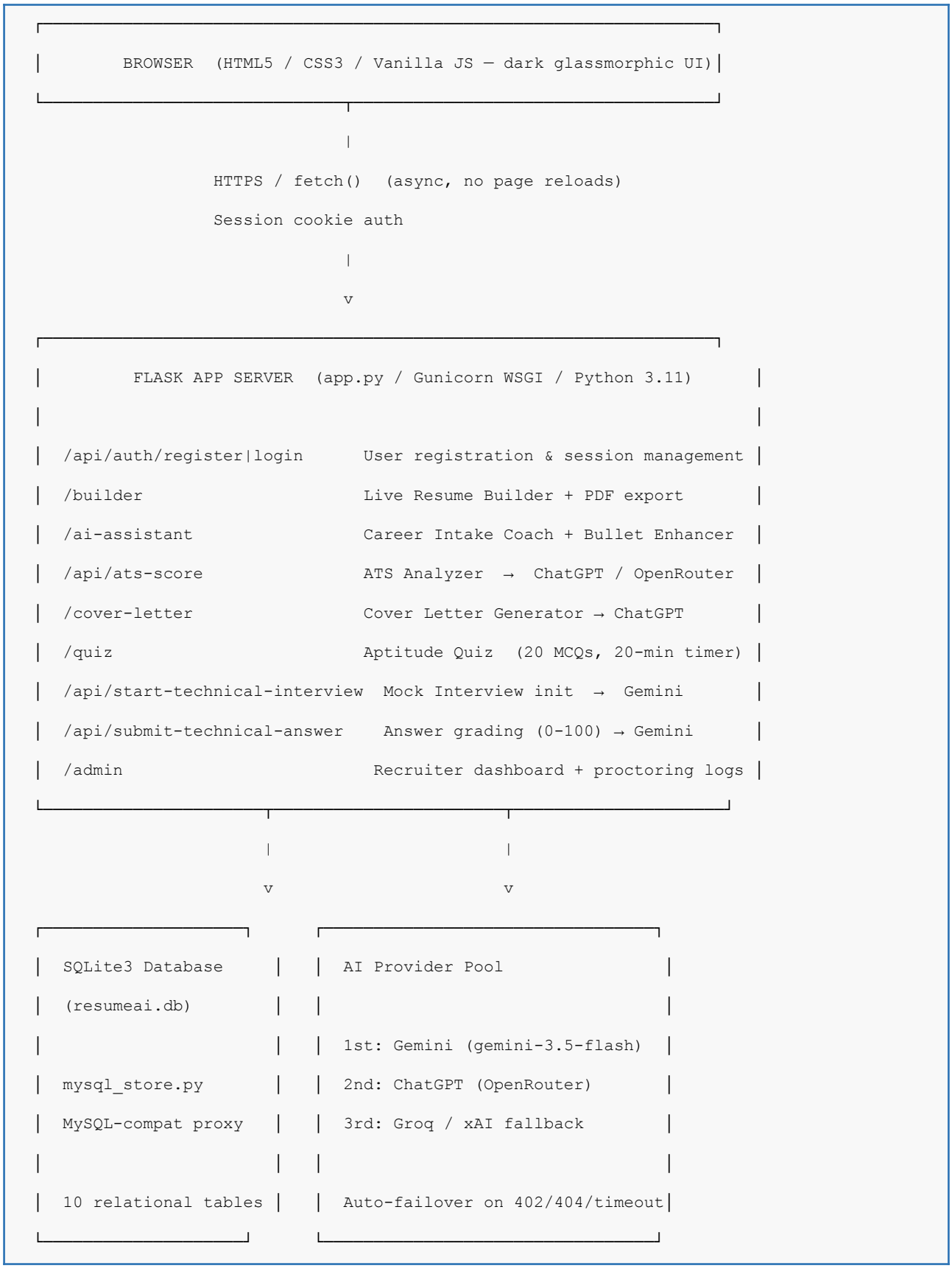
Phase	Focus	Key Activities	Key Artifacts
Phase 1 Requirements & Specification	Define scope, user roles, and dual-LLM routing strategy	Requirements gathering; dual-LLM routing decision; 10-table ER diagram; API endpoint map	Feature list, ERD draft, API spec
Phase 2 Database & Auth	SQLite schema and secure authentication layer	10-table schema with cascade deletes; pbkdf2:sha256 password hashing; session token management via mysql_store proxy	database.py, mysql_store.py, users / user_sessions tables
Phase 3 Frontend UI Modules	Responsive dark glassmorphic UI across all seven modules	Split-screen resume builder; ATS analyzer form; quiz countdown UI; AI chat interface; recruiter admin dashboard	HTML/CSS/JS templates; shared CSS design-token variables
Phase 4 Flask API & LLM Pool	Backend routing and dual-provider AI integration	REST endpoints for auth, ATS, interview, quiz, cover letter; Gemini + ChatGPT/OpenRouter fallback chain with automatic failover	app.py routes, AI provider pool module
Phase 5 E2E Integration Testing	Full-stack validation and bug resolution	E2E test client: registration → ATS → quiz → interview flow; proctoring log validation; PDF export cross-browser testing; API 404 model endpoint fix	Test scripts, bug-fix commits, updated model config
Phase 6 Cloud Deployment	Production deployment on Render with secrets management	Gunicorn WSGI config; SQLite path fix for Render ephemeral disk; API key secrets via Render Environment Variables panel; max_tokens cap to prevent 402 billing blocks	Live URL: http://13.53.172.71/

This sequencing — schema and auth first, UI second, AI routing third, deployment last — reflects an iterative approach where each layer was built and validated against a working layer beneath it. Configuration corrections (deprecated model endpoint, SQLite path, token cap) were discovered during Phase 5 integration testing and resolved before deployment.

4. Application Architecture

ResumeAI follows a three-tier architecture: a Flask monolith acts as secure middleware between the browser client and two external service layers — an embedded SQLite relational database for persistence, and the dual-LLM provider pool (Gemini primary, ChatGPT/OpenRouter secondary) for AI inference. A mysql_store.py adapter wraps the SQLite engine behind a MySQL-compatible interface, decoupling all application routes from the database engine so a future migration to hosted MySQL or PostgreSQL requires only a connector swap.




BROWSER (HTML5 / CSS3 / Vanilla JS — dark glassmorphic UI)
HTTPS / fetch() (async, no page reloads)
Session cookie auth
FLASK APP SERVER (app.py / Gunicorn WSGI / Python 3.11)
/api/auth/register|login User registration & session management
/builder Live Resume Builder + PDF export
/ai-assistant Career Intake Coach + Bullet Enhancer
/api/ats-score ATS Analyzer → ChatGPT / OpenRouter
/cover-letter Cover Letter Generator → ChatGPT
/quiz Aptitude Quiz (20 MCQs, 20-min timer)
/api/start-technical-interview Mock Interview init → Gemini
/api/submit-technical-answer Answer grading (0-100) → Gemini
/admin Recruiter dashboard + proctoring logs
SQLite3 Database
(resumeai.db)
mysql_store.py
MySQL-compat proxy
10 relational tables
AI Provider Pool
1st: Gemini (gemini-3.5-flash)
2nd: ChatGPT (OpenRouter)
3rd: Groq / xAI fallback
Auto-failover on 402/404/timeout

4.1 Database Design

All tables use SQLite3 with ON DELETE CASCADE constraints on foreign keys to maintain referential integrity. The mysql_store.py adapter emulates MySQL endpoint behaviour, decoupling every Flask route from the underlying database engine.

users	id, full_name, email (UNIQUE), password_hash (pbkdf2:sha256), created_at, last_login_at
user_sessions	id, user_id (FK → users, CASCADE), token_hash (SHA-256, UNIQUE), expires_at, created_at
resumes	id, filename, content (full extracted text), created_at
candidates	id, name, email (UNIQUE), phone, created_at
interview_sessions	id (UUID PK), candidate_id (FK), resume_id (FK), job_role, resume_score, status CHECK('started'|'completed'|'terminated'), duration_minutes, created_at, completed_at
technical_answers	id, session_id (FK, CASCADE), question_text, candidate_answer, score CHECK(0–100), feedback, created_at
aptitude_answers	id, session_id (FK, CASCADE), question_text, candidate_answer, is_correct BOOLEAN(0/1), created_at
proctoring_logs	id, session_id (FK, CASCADE), event_type, timestamp, details — logs tab-switch and window-blur events for recruiter review
final_reports	id, session_id (FK UNIQUE, CASCADE), score CHECK(0–100), summary, technical_rating, communication_rating, created_at
learning_roadmaps	id, session_id (FK UNIQUE, CASCADE), plan_json — 4-week study plan with curated free-course links (Harvard CS50, MIT OCW)

4.2 Key API Endpoints

Endpoint	Method	Purpose	Notes
/api/auth/register	POST	Register a new candidate account	HTTP 400 for duplicate email or password < 8 chars
/api/auth/login	POST	Authenticate and issue session cookie	HTTP 401 for invalid credentials
/api/ats-score	POST	Upload resume PDF + paste JD; returns Compatibility Score	PDF capped at 5 MB; NLP comparison via ChatGPT/ OpenRouter
/api/start-technical-interview	POST	Create session UUID and generate role-specific questions	Returns session ID; requires candidate profile
/api/submit-technical-answer	POST	Grade candidate answer; return score (0–100) + feedback	Graded by Gemini; persisted to technical_answers table
/api/cover-letter	POST	Generate formatted cover letter from resume + target role	Output returned as editable text block
/admin	GET	Recruiter dashboard — profiles, scores, proctoring logs	Admin role required; AI hiring summary per candidate

5. Challenges Encountered and How They Were Resolved

Challenge	Root Cause	Solution Applied	Key Learning
API 404 — Deprecated Model Endpoint	Google deprecated the gemini-1.5-flash endpoint; all requests to the old identifier began returning 404	Updated the active model string to gemini-3.5-flash in the central config file; confirmed resolution with a test call before redeployment	Pin model identifiers in a single config file, not scattered across route code. Monitor provider changelogs proactively.
Render SQLite Data Reset on Redeploy	Render Free Tier containers use ephemeral disks that are wiped on every restart or redeploy, erasing the SQLite database file	Added a directory-creator bootstrap routine in the SQLite connector to recreate the schema on startup; documented the limitation with instructions for attaching a Render Persistent Disk or migrating to an external DB in production	Cloud containers are stateless by default. Any file written to the container filesystem is ephemeral unless a persistent volume is explicitly mounted.
OpenRouter 402 — Free-Tier Credit Block	API request payloads did not specify a token limit, causing requests to consume more tokens than the free-tier credit balance allowed	Added max_tokens: 1000 to all API request payloads as an explicit ceiling; restructured prompts to be more concise without losing instruction fidelity	Token budgeting is a first-class reliability concern, not just a cost concern — an unbounded call that hits a 402 silently breaks a feature for the end user.
Windows Terminal Unicode Crash	Unicode checkmark characters (✔) in test log output caused encoding errors in terminals using CodePage 1252 (Windows default)	Replaced all unicode symbols in test log strings with ASCII-safe alternatives such as [OK] and [FAIL]	Cross-platform terminal output must use only ASCII-safe characters. Emoji and unicode symbols should be confined to UI layers, not server-side logs.
PDF Resume Export Formatting Inconsistency	HTML/CSS canvas rendered correctly on screen but did not translate 1:1 to PDF across different browsers via the native print API	Implemented a dedicated print stylesheet with explicit @page rules, controlled page-break CSS properties, and fixed margin definitions for the PDF export pipeline	Browser-native PDF export requires a purpose-built print stylesheet — screen CSS alone is insufficient for reliable cross-browser document rendering.

6. Key Learning & Reflection

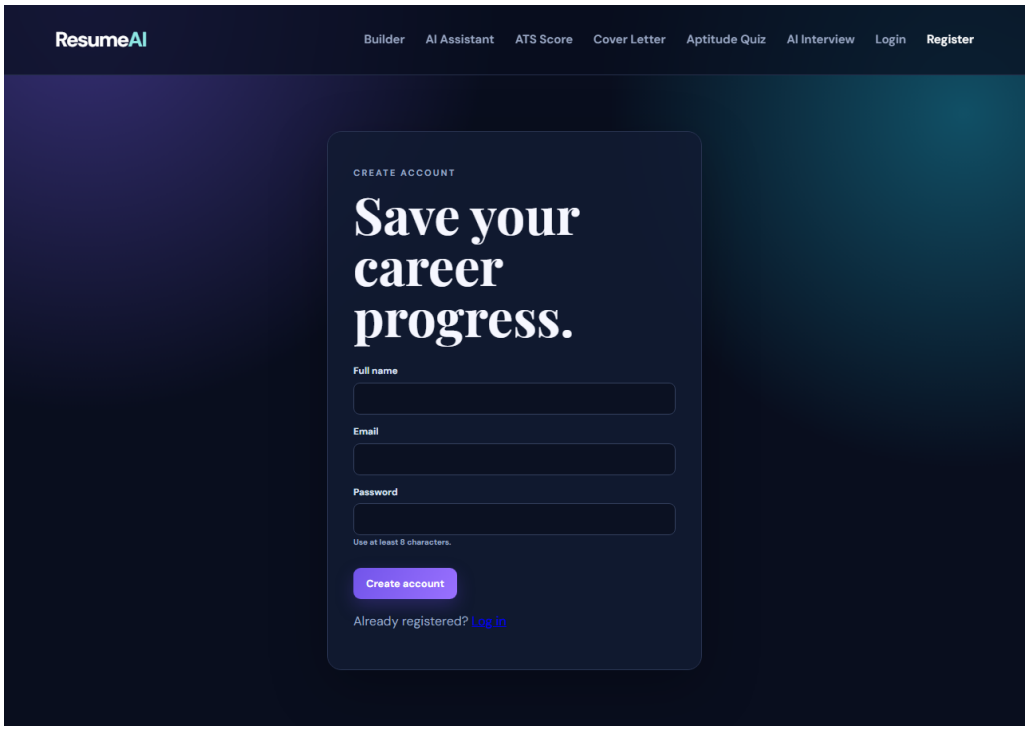
- **Dual-LLM routing is worth the added complexity.** Assigning Gemini to speed-critical conversational modules and ChatGPT to precision-critical semantic tasks produced measurably better outputs than a single-model approach. The fallback chain meant no module went dark when any provider hit a rate limit.
- **Database adapter layers pay dividends early.** Writing all Flask routes against the mysql_store MySQL-compatible interface, rather than calling SQLite directly, was a minor upfront investment. It means the entire data layer can be migrated to a hosted production database by swapping a single connector — no route rewrites required.
- **Profile context must be threaded deliberately across AI calls.** Because every LLM call is stateless, carrying the candidate's resume text and target role into every relevant prompt (ATS, cover letter, mock interview) required explicit prompt engineering for each module. The payoff was that each module felt genuinely personalised rather than generic — a key differentiator from disconnected point tools.
- **Cloud containers are ephemeral by default.** The Render SQLite reset forced a proper solution: a schema-recreation bootstrap and documented guidance for attaching persistent storage. The lesson applies to any containerised deployment — never assume the local filesystem persists across restarts.
- **Token budgeting is a reliability concern, not just a cost concern.** An API call with no token ceiling that hits a 402 fails silently from the user's perspective. Explicit max_tokens caps on every request are as important as error handling for maintaining a stable user experience on free-tier infrastructure.
- **Browser focus APIs are a lightweight but effective proctoring layer.** Monitoring tab-switch and window-blur events client-side and persisting violations to the proctoring_logs table gave recruiters meaningful anti-cheat data with almost no backend overhead — a good example of combining simple browser primitives with database persistence to create a feature that feels more sophisticated than its implementation.

7. Live Application

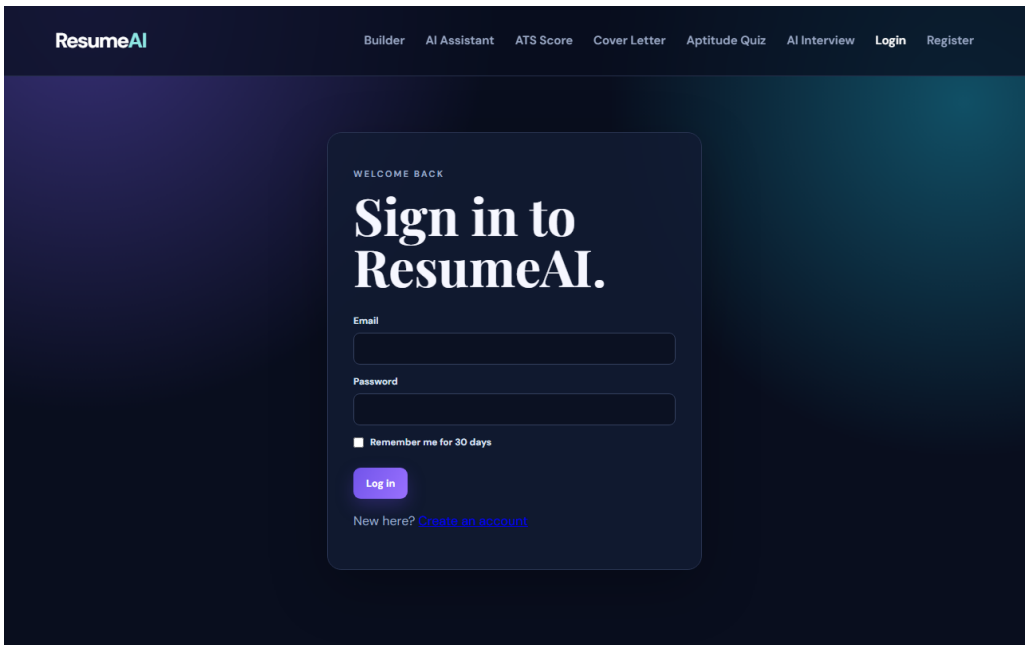
The application is deployed and publicly accessible at:

Live Application: https://ai-resume-analyzer-and-ai-interview.onrender.com/

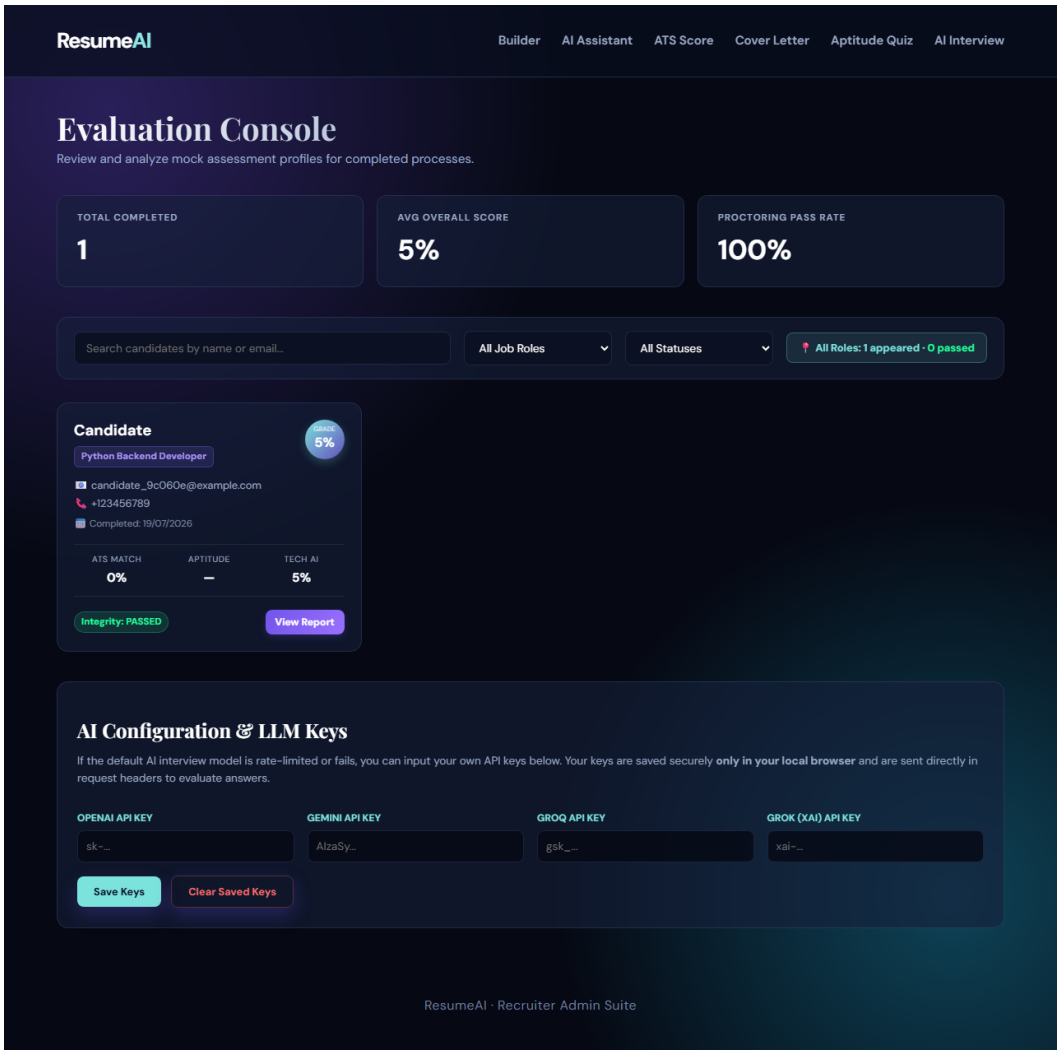
Register Page



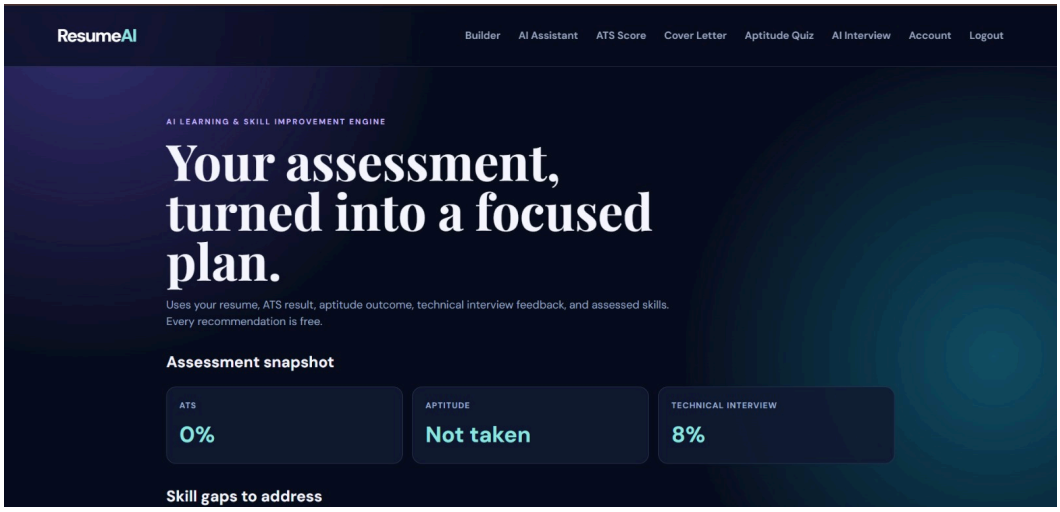
Login Page

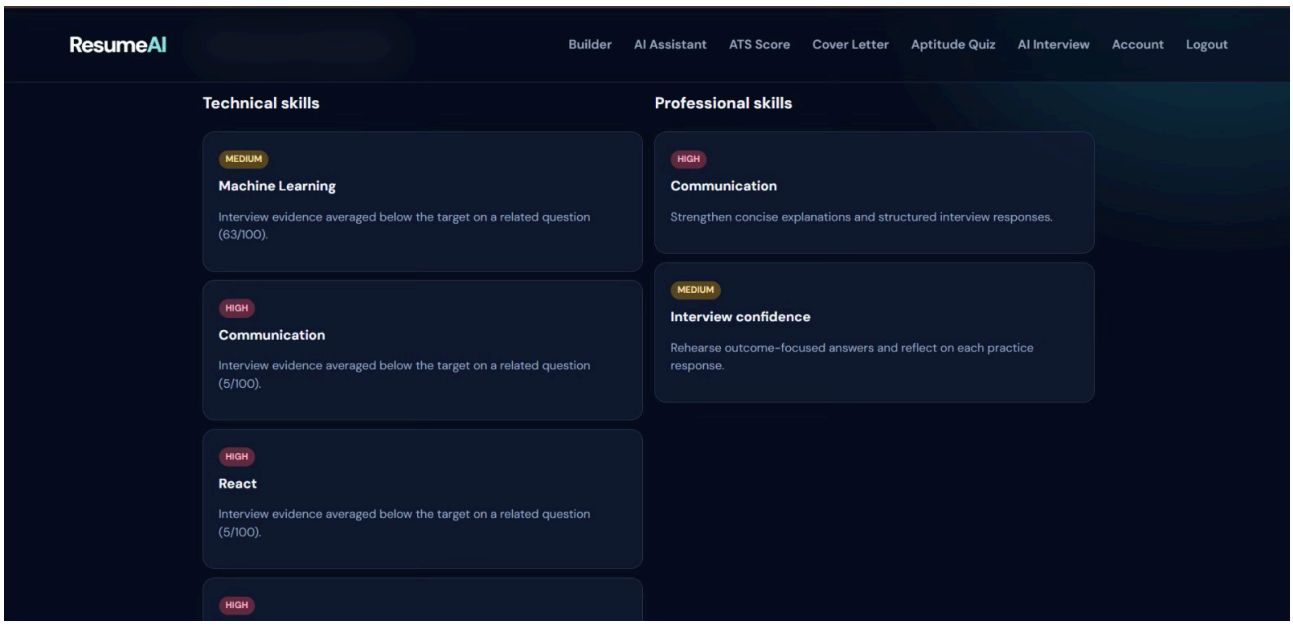


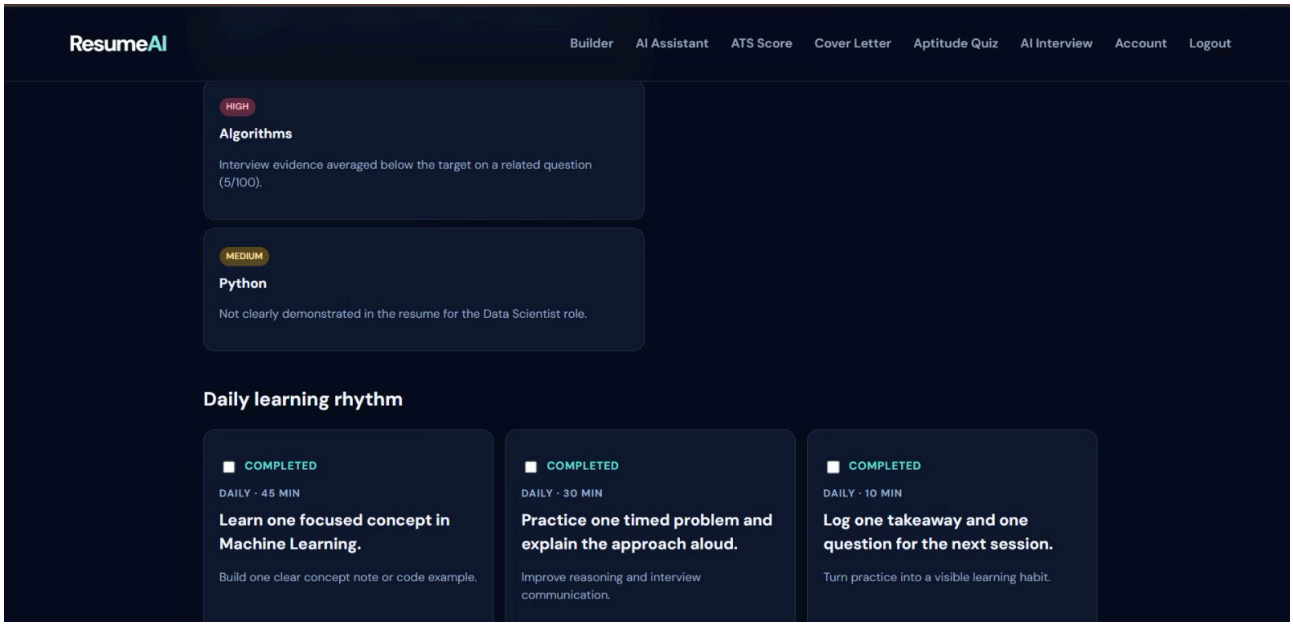
Admin Page

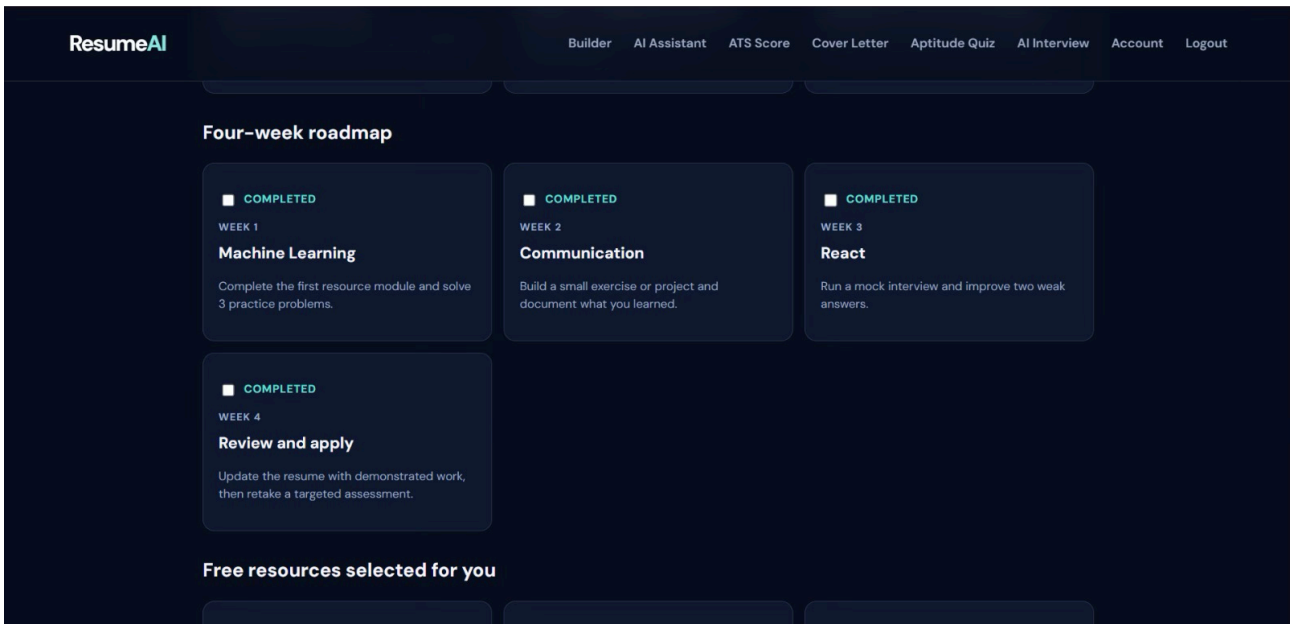


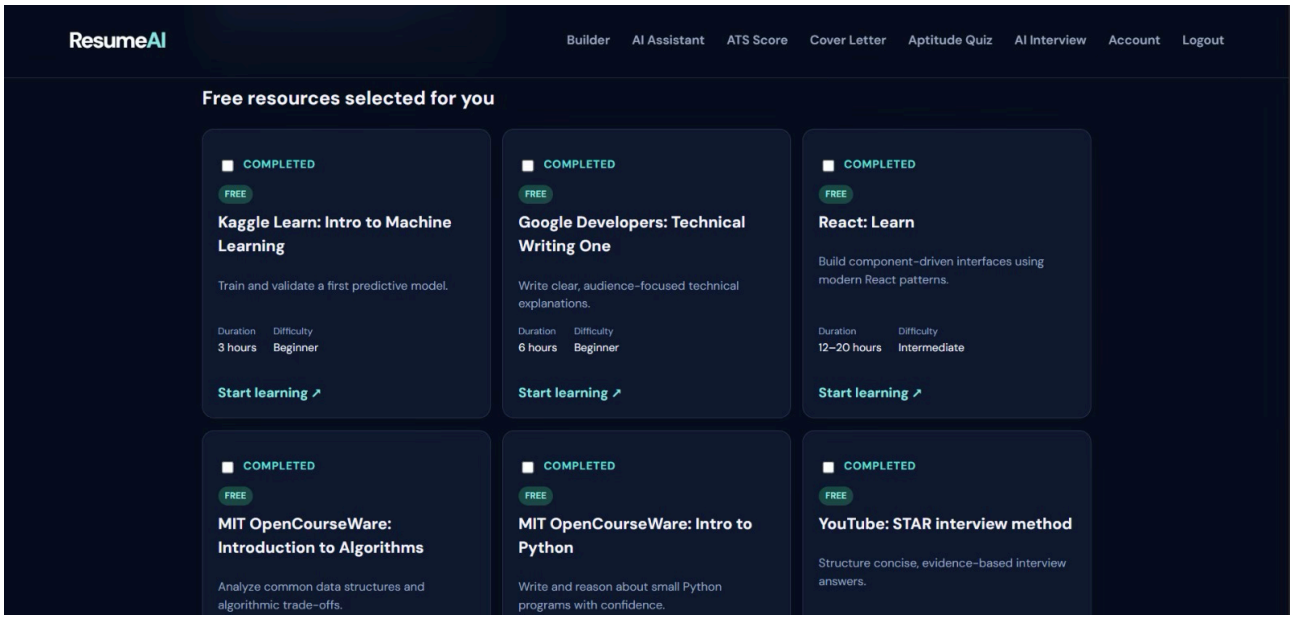
Learning Plan



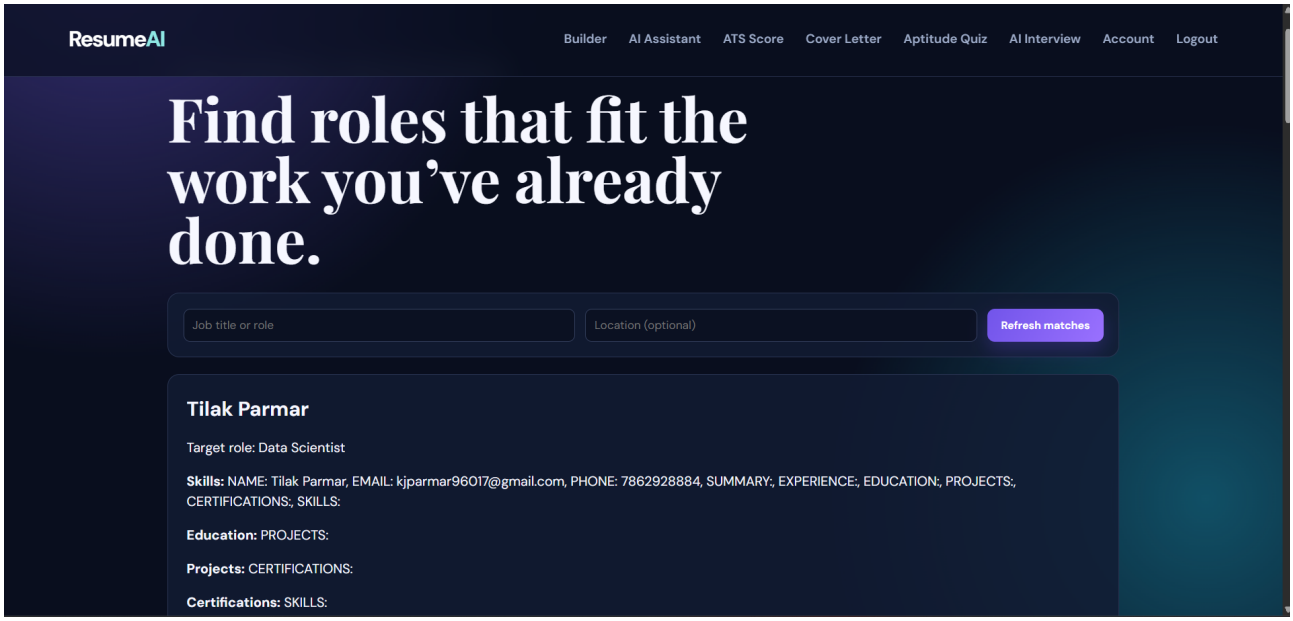

ResumeAI
Builder
AI Assistant
ATS Score
Cover Letter
Aptitude Quiz
AI Interview
Account
Logout
Technical skills
MEDIUM
Machine Learning
Interview evidence averaged below the target on a related question (63/100).
HIGH
Communication
Interview evidence averaged below the target on a related question (5/100).
HIGH
React
Interview evidence averaged below the target on a related question (5/100).
HIGH
Professional skills
HIGH
Communication
Strengthen concise explanations and structured interview responses.
MEDIUM
Interview confidence
Rehearse outcome-focused answers and reflect on each practice response.

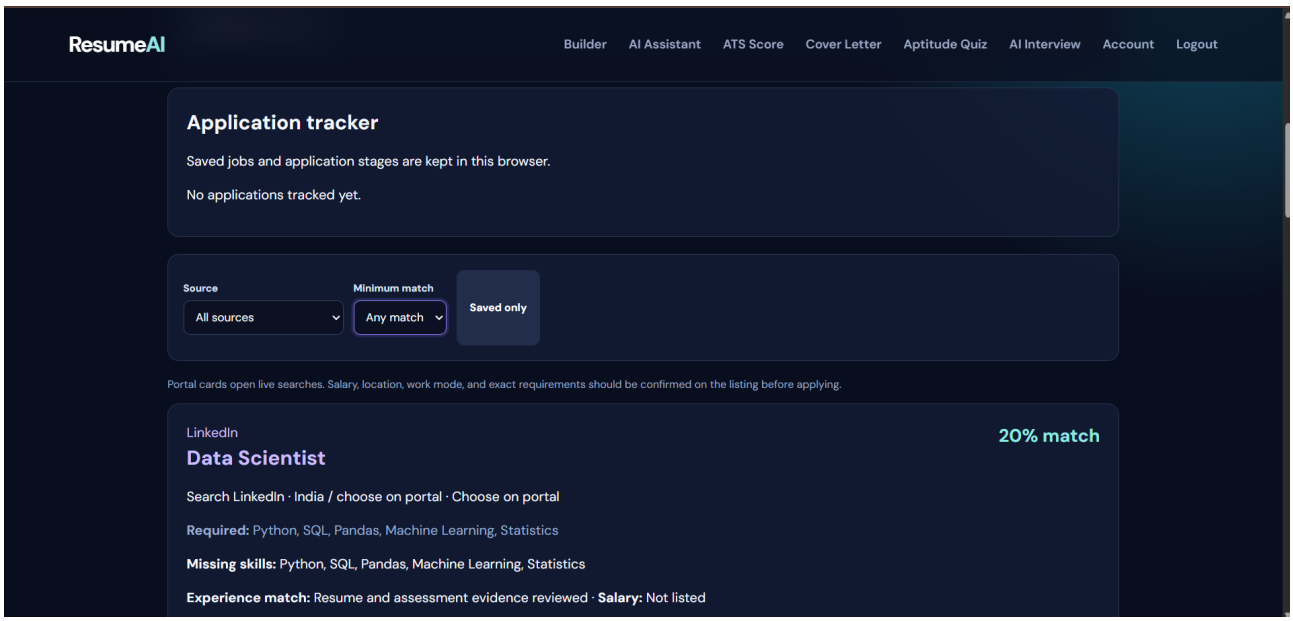

ResumeAI
Builder
AI Assistant
ATS Score
Cover Letter
Aptitude Quiz
AI Interview
Account
Logout
HIGH
Algorithms
Interview evidence averaged below the target on a related question (5/100).
MEDIUM
Python
Not clearly demonstrated in the resume for the Data Scientist role.
Daily learning rhythm
COMPLETED
DAILY · 45 MIN
Learn one focused concept in Machine Learning.
Build one clear concept note or code example.
COMPLETED
DAILY · 30 MIN
Practice one timed problem and explain the approach aloud.
Improve reasoning and interview communication.
COMPLETED
DAILY · 10 MIN
Log one takeaway and one question for the next session.
Turn practice into a visible learning habit.

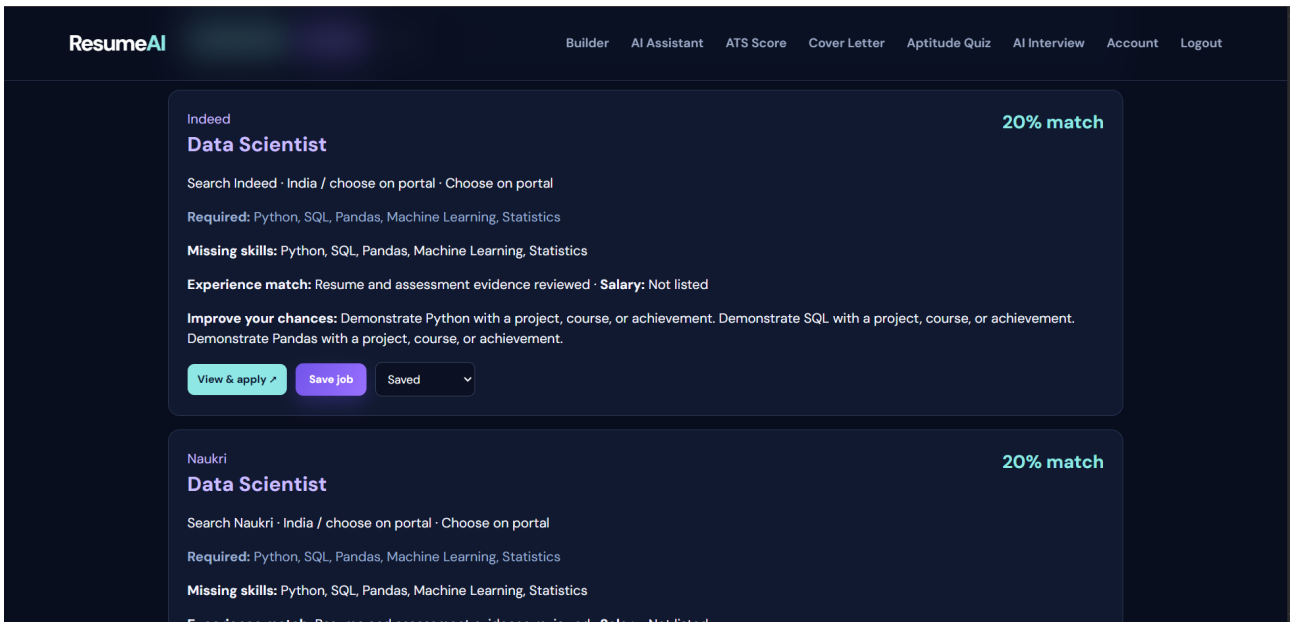

ResumeAI
Builder
AI Assistant
ATS Score
Cover Letter
Aptitude Quiz
AI Interview
Account
Logout
Four-week roadmap
COMPLETED
WEEK 1
Machine Learning
Complete the first resource module and solve 3 practice problems.
COMPLETED
WEEK 2
Communication
Build a small exercise or project and document what you learned.
COMPLETED
WEEK 3
React
Run a mock interview and improve two weak answers.
COMPLETED
WEEK 4
Review and apply
Update the resume with demonstrated work, then retake a targeted assessment.
Free resources selected for you



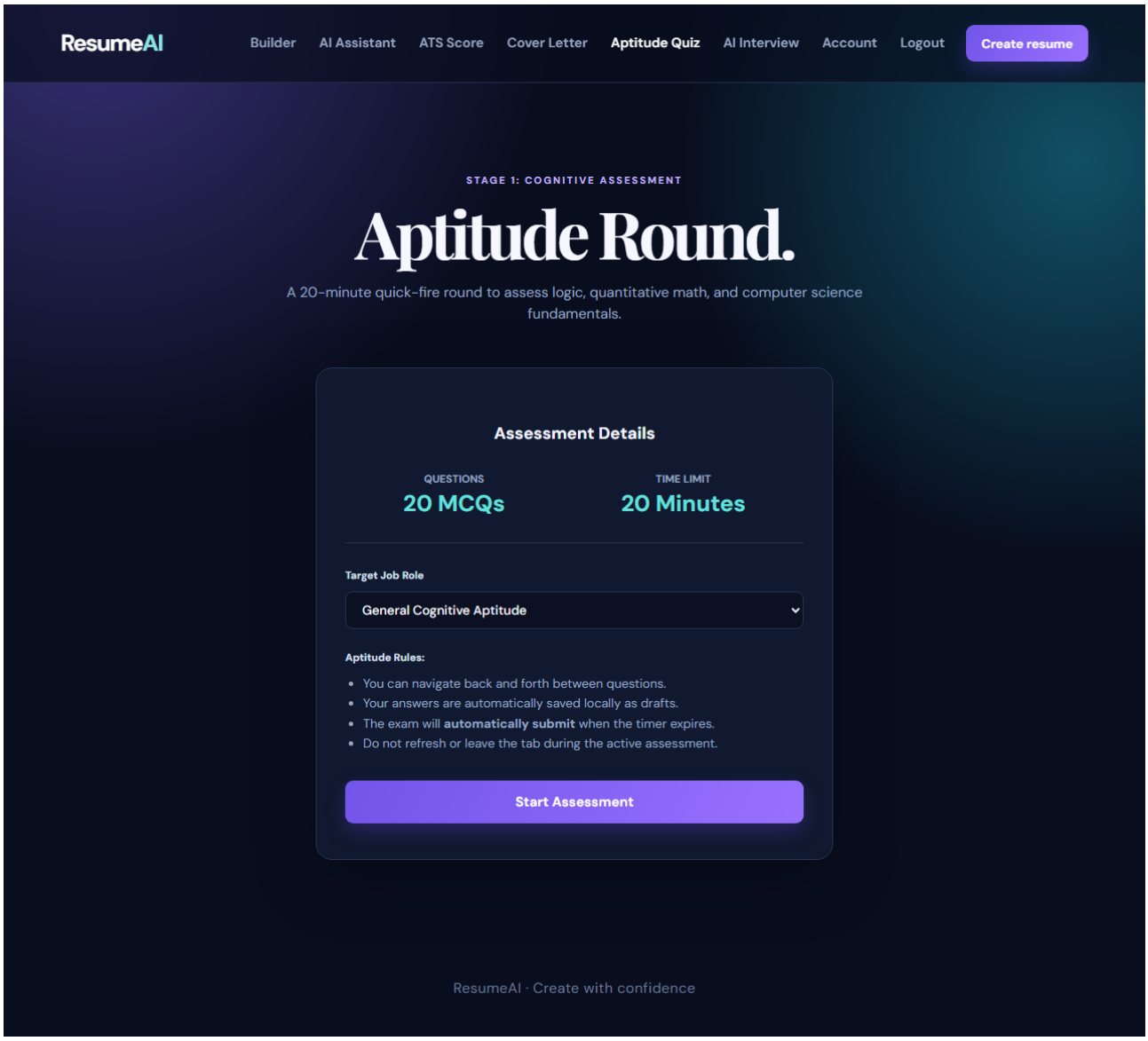
Job Discovery



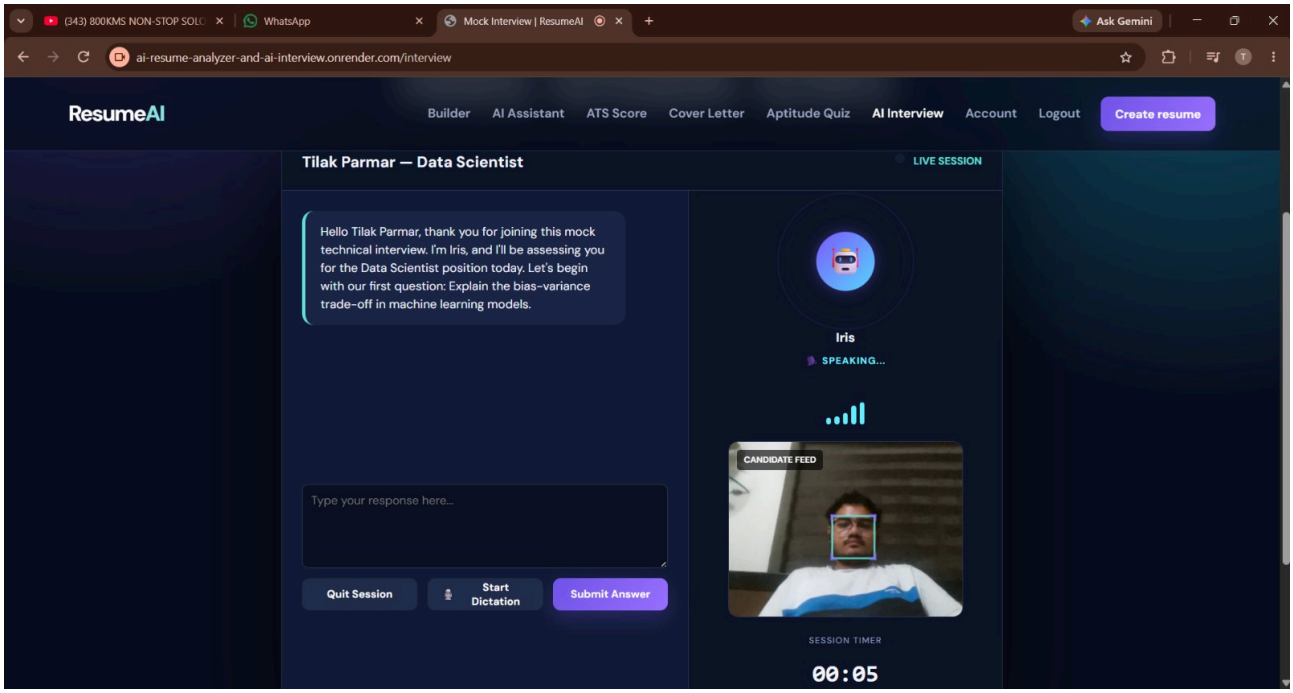




Aptitude Round



Live Interview



- **Home / Landing Page —** entry point with module navigation and registration prompt.
- **User Dashboard (My Career Space) —** saved resume drafts, job-application tracker, and past interview session history.
- **Live Resume Builder —** split-screen input / live preview canvas; template and accent colour selector; one-click PDF export.
- **Dual-Mode AI Assistant —** Career Intake Coach conversational panel and Bullet Enhancer text-transformation interface.
- **ATS Compatibility Analyzer —** resume PDF upload + job description input; Compatibility Score with Matched Skills / Missing Skills breakdown.
- **Aptitude Quiz Simulator —** 20-question timed exam with countdown, autosave, auto-submit, and anti-cheat tab warnings.
- **Adaptive AI Mock Interview —** conversational hiring-manager interface with real-time question adaptation, pacing timer, and background proctoring.
- **Recruiter / Admin Dashboard —** candidate dossiers, ATS scores, aptitude results, AI-generated qualitative hiring summaries, and proctoring violation logs.